Table 1: **Ablation on selection strategies under a fixed compute budget.** We evaluate DGS, Random-k, Bottom-k, and a Top/Bottom Mix on a representative concept subset (one human, one animal, and one object), retaining $k = 9$ trajectories for policy updates and using an expanded candidate pool of size $N = 36$. Acceleration ratios are reported relative to the Sync-GRPO baseline. *Ref.* denotes a reference setting not directly comparable in efficiency, and *Not reached* indicates failure to attain 98% of the baseline peak reward within the maximum training budget.

Selection Strategy	**Rea.***	**Dense Rea.***	**Dense Gen.***	**Rea. Gen.***	**Dense Rea. Gen.***	**Acceleration Ratio**
Sync-GRPO	0.269	0.531	0.486	0.571	0.578	1.0×
Full-N Oracle	0.286	0.537	0.527	0.611	0.599	Ref.
Random-k	0.270	0.533	0.483	0.578	0.577	0.9×
Bottom-k	0.264	0.489	0.469	0.555	0.539	Not reached
Top/Bottom Mix	0.271	0.524	0.477	0.569	0.582	0.7×
Sync-GRPO+DGS	0.281	0.536	0.503	0.587	0.583	1.8×

Table 2: **Ablation on the $\alpha_{\text{text}}/\alpha_{\text{image}}$ weighting schedule.** We vary the relative weighting between the text-side and image-side policy optimization terms while keeping all other settings fixed.

$\alpha_{\text{text}} : \alpha_{\text{image}}$	**Rea.***	**Dense Rea.***	**Rea. Gen.***		**Dense Rea. Gen.***	
	BLEU	**GPT**	**CLIP-T**	**CLIP-I**	**GPT**	**CLIP-I**
0.2:0.8	0.221	0.439	0.271	0.797	0.338	0.752
0.4:0.6	**0.250**	**0.503**	**0.324**	**0.801**	**0.353**	**0.756**
0.6:0.4	0.256	0.507	0.319	0.791	0.361	0.747
0.8:0.2	0.243	0.501	0.295	0.732	0.354	0.718

Figure 1: **What does DGS actually discard?** For three representative concepts, we compare the reference image, a trajectory discarded by DGS, a retained low-quality trajectory, and a retained high-quality trajectory. Discarded samples are typically catastrophic structural failures, whereas retained low-quality samples remain visually plausible but lack fine-grained details. This suggests that DGS primarily filters low-information noise rather than removing informative hard negatives.